

# RAJ KUMAR

## Lead Full Stack Engineer & Systems Architect

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### PROFESSIONAL SUMMARY

High-impact **Lead Full Stack & Systems Engineer** with 6+ years of specialized experience architecting and scaling enterprise distributed backends, compliance platforms, and modern web/mobile applications. Proficient in **TypeScript, Python, Node.js, NestJS, FastAPI, and React**, with proven expertise building mission-critical integrations: Australian Government **PRODA/CCS compliance**, multi-tenant **X12 EDI & AS2 cryptographic transports**, **Stripe Connect/BECS Direct Debit billing**, real-time **Socket.IO** clusters, and **OpenAI/LLM-powered candidate matching pipelines**. Passionate about modular micro-services, double-entry transactional ledgers, robust asynchronous job queues (BullMQ/ARQ), and production-grade cloud reliability.

### TECHNICAL COMPETENCIES

|                                    |                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>                  | TypeScript, JavaScript (ES6+), Python 3.12, PHP, Dart, Java, SQL, HTML5/CSS3                                     |
| <b>Backend &amp; Frameworks:</b>   | NestJS 11, FastAPI (Async), Node.js, AdonisJS v6, Express.js, Laravel, REST APIs, WebSockets, Socket.IO          |
| <b>Frontend &amp; Mobile:</b>      | React 18, Vite, Next.js, React Native (iOS & Android), TanStack Query, Tailwind CSS, shadcn/ui, Bootstrap        |
| <b>AI &amp; Data Pipelines:</b>    | OpenAI SDK, Prompt Engineering, Resume/Document Ingestion (pdf-parse/mammoth), Computer Vision (OpenCV, K-Means) |
| <b>Protocols &amp; Transports:</b> | X12 EDI (850, 945, 944, 810, 997, 855), AS2 (OpenSSL CMS Encrypt/Sign, sync/async MDN), SFTP, S3/SQS             |
| <b>Databases &amp; Queues:</b>     | PostgreSQL (asyncpg/Neon), MySQL, MongoDB/Mongoose, Redis 7, BullMQ, ARQ, TypeORM, Drizzle, SQLAlchemy 2.0       |
| <b>Integrations &amp; Billing:</b> | Stripe (Connect Express, BECS Direct Debit, Billing, Elements), PRODA/CCS Gov API, Xero Payroll, Mapbox GL       |
| <b>DevOps &amp; Security:</b>      | Docker, Docker Compose, AWS (S3, SQS, Secrets Manager), Azure Blob, JWT + OAuth2 rotation, TOTP 2FA, Git, CI/CD  |

### PROFESSIONAL EXPERIENCE

#### Associate Software Engineer / Lead Full Stack Engineer — Luminoguru Pvt Ltd

2020 – Present | Mohali, India

- Architect and engineer full-stack platforms, high-throughput asynchronous backend services, and scalable web/mobile applications.
- Engineered the flagship TGA childcare compliance engine (56 entities, 34 controllers) interfacing with Australian Government PRODA cryptographic auth and automated Stripe AU BECS recurring billing.
- Built enterprise multi-tenant X12 EDI middleware in Python 3.12/FastAPI handling AS2 encrypted communications, idempotent deduplication, and ERP sync.
- Designed and implemented full-stack AI recruitment platform (Talent Bridge) integrating OpenAI SDK for automated candidate scoring and resume parsing.
- Spearheaded real-time architectures with Socket.IO and BullMQ, handling high-frequency chats, live location maps, and workforce event scheduling.

#### Web Developer — Quadrish Innovations

2019 – 2020 | Chandigarh, India

- Developed scalable database-driven web platforms using PHP, CodeIgniter, Laravel, and MySQL following strict OOP patterns.
- Engineered RESTful APIs, optimized complex relational queries, and introduced Node.js/MongoDB micro-modules for real-time throughput.

#### Trainee / Android & Web Developer — Tuespot Pvt Ltd

2017 – 2019 | Mohali, India

- Developed native Android applications using Java, XML, and SQLite, implementing responsive UI components and background sync.
- Collaborated on dynamic web portal development utilizing PHP, JavaScript, HTML5, CSS3, and relational database schema design.

### HIGH-PROFILE FEATURED PROJECTS

#### TGA Grove Academy — Childcare Management & Gov Compliance Platform

[tga.edu.au](https://tga.edu.au)

Stack: TypeScript, NestJS 11, PostgreSQL, TypeORM, React 18, Vite, Tailwind, PRODA/CCS, Stripe BECS, Puppeteer

- Architected enterprise childcare management system (500+ files, 34 controllers, 56 entities) with role-scoped staff/parent portals and kiosk PIN attendance.
- Engineered end-to-end Australian CCS compliance pipeline via PRODA device auth/cryptographic key rotation, automated session-reporting state machine (INIT/VARY/NOCHG/NO CARE), and audit trails.
- Implemented automated direct-debit billing using Stripe AU BECS with webhook idempotency, retry scheduling, and automated PDF invoice generation via Puppeteer.

#### EDI Middleware — Multi-Tenant X12 Transport & ERP Integration

Production Enterprise System

Stack: Python 3.12, FastAPI (Async), PostgreSQL 16, SQLAlchemy 2.0, Redis 7, ARQ Workers, AS2 (OpenSSL CMS), pyx12, AWS SQS/S3, Docker

- Designed async multi-tenant X12 pipeline processing inbound 850/945/944/810 documents and generating outbound 997/855/940 acknowledgments with status tracking.
- Implemented production AS2 transport: OpenSSL CMS signing/encryption, partner certificate store, sync/async MDN handling, and integrated live warehouse partner (Solvira).
- Engineered resilient ingestion engine with Redis + ARQ workers, PostgreSQL audit trail with control-number deduplication (409 handling), and swappable REST ERP adapter (Icicle Foods).

## Talent Bridge — AI-Powered Recruitment & Candidate Matching Platform

[Full-Stack AI Monorepo](#)

Stack: TypeScript, NestJS 11, PostgreSQL, TypeORM, Drizzle ORM, OpenAI SDK, Redis, BullMQ, React 18, Vite, TanStack Query, Tailwind, Stripe

- Built full-stack AI recruitment platform: automated resume extraction (PDF/DOCX), OpenAI-powered scoring, and asynchronous candidate-job matching via BullMQ worker queues.
- Shipped multi-role architecture (Admin, Recruiter, Enterprise) with Stripe subscription tiers, checkout sessions, team invitations, and comprehensive Recharts analytics.
- Engineered responsive 3-portal SPA with TanStack Query caching, form validations (RHF + Zod), inline PDF resume viewing, and AWS S3 presigned asset storage.

## Globetrottr — Travel Social Network & Real-Time Messaging Platform

[globetrottrapi.lusites.xyz](https://globetrottrapi.lusites.xyz)

Stack: TypeScript, NestJS 11, MongoDB, Mongoose 8, Redis, BullMQ, Socket.IO, Mapbox GL, Elasticsearch, Typesense, AWS S3, Stripe

- Architected large-scale social backend (~338 TS files, 29 controllers, 60 services) powering social travel feeds, stories, country passport tracking, and dynamic Mapbox clustering.
- Built high-concurrency real-time messaging gateway using Socket.IO with typing indicators, presence tracking, media sharing via AWS S3, and Google Cloud automated translation.
- Integrated Tiqets Distributor API for travel event discovery, business accounts with QR redemption, and Stripe subscriptions.

## ADDITIONAL PRODUCTION PROJECTS

Extensive portfolio of 15+ production systems, mobile apps, and AI engines detailed on my website: <https://rajansandha.github.io/dev-portfolio>

- **Givicle:** Full-stack CSR employee-giving platform with double-entry contribution ledger, payroll deductions, and statement exports. (AdonisJS v6, React 18, MySQL, Redis, BullMQ, Azure Blob, S3)
- **BarChase API:** Nightlife discovery platform with Mapbox geo-filtering, crowd status tracking, Stripe sponsorships, and push notifications. (NestJS 11, MySQL, TypeORM, Stripe, Firebase)
- **TGA-Mesh:** Childcare workforce ops platform with shift rostering, timesheet aggregation, Xero payroll sync, and GPS attendance. (React Native, NestJS, MongoDB, Socket.IO, Xero API)
- **Village Vest:** Community group-funding escrow platform with Stripe Connect Express onboarding, OFAC compliance screening, and SMS OTP. (NestJS 11, React 18, Stripe Connect, Twilio)
- **StenGen:** AI image-to-stencil conversion pipeline utilizing K-Means clustering and automated vector SVG generation. (Python, OpenCV, NestJS, SVG Engine)
- **Boldify:** High-volume Shopify brand storefront engineered with custom modular Liquid snippet architecture. (Shopify Liquid, App Integrations)
- **Mobile & AI Labs:** Published mobile applications (ChessRoyal Game on Google Play) and active generative AI experiments (FashionTryOn virtual fitting room & ContentCreationAI voice cloning).

## EDUCATION

### Master of Computer Applications (MCA)

2016 – 2018 | Grade A

Maharaja Ranjeet Singh Punjab Technical University (MRSPTU), Punjab, India

### Bachelor of Computer Applications (BCA)

2012 – 2015 | Grade A (65%)

Panjab University, Chandigarh, India